ZI BUS

How it works

A plain-language guide to the ZI Bus platform: passengers, bus operators, bus crew, highway dhabas and the ZI Bus team.

Who uses ZI Bus
Five groups of people and what each one does

A passenger's trip
From search to getting off the bus

Seats & other platforms
Why a seat is never sold twice

Dhaba food
How food is ordered, cooked and served on the way

Money, readiness, glossary
Who gets paid what, and what is left before launch

Prepared October 2026 | Status: built and tested, pending launch setup

SECTION 1

The big picture

ZI Bus is our own bus-booking platform, like RedBus but under our brand. We own the software, the database and the booking engine. We do not borrow bus seats from anyone else's system: bus companies put their buses on ZI Bus, and ZI Bus keeps the one true record of which seat is free and which is sold.

On top of tickets, ZI Bus adds something other platforms do not: passengers can pre-order food from highway dhabas along the route, and the dhaba is told to start cooking about 30 minutes before the bus arrives.

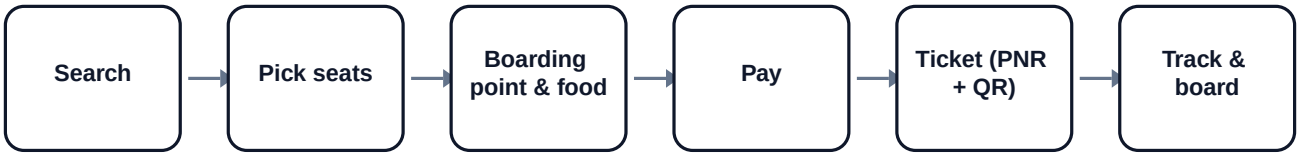
> **IN ONE SENTENCE**
> Passengers book seats and food in one app, bus companies manage their buses and earnings, dhabas get food orders on time, and ZI Bus keeps every seat count correct across all the places a bus is sold.

Who uses ZI Bus

Who	Where they use it	What they do
Passengers	ZI Bus app	Search buses, pick seats, choose where to get on and off, add dhaba food, pay, get a ticket with a QR code, track the bus live.
Bus operators (companies)	Operator panel	Add buses, routes, timings, fares and seat layouts. See bookings, passenger lists and what they will be paid.
Bus crew (driver / conductor)	Driver app	Start the trip so passengers can track it. Check passengers in at the door by scanning their QR or typing the PNR.
Highway dhabas	Dhaba panel (website)	Keep their menu and prices up to date. Get an alert before each bus arrives. Mark food as cooking, ready and served.
ZI Bus team	Admin panel	Approve and manage operators, buses and dhabas. Connect other booking platforms. Watch seat counts, food orders and commission.

SECTION 2

A passenger's trip, step by step



1

Search

The passenger enters From, To and the date. ZI Bus lists the buses with timings, prices, AC or non-AC, sleeper or seater, and how many seats are left.

2

Pick seats on a live seat map

The seat map shows the lower and upper deck exactly as the bus is laid out. Grey seats are taken. The map updates by itself: if someone else books a seat (on ZI Bus or on another platform) while the passenger is looking, that seat turns grey immediately, and if they had selected it, they are told politely and it is removed from their selection.

3

Choose where to get on and off

The passenger picks a boarding point (for example "Swargate, 21:00") and a dropping point from the stops the operator has set for that bus.

4

Add food (optional)

If the bus stops at partner dhabas, their menus appear with the time the bus is expected there. The passenger adds dishes; the price is added to the same payment. Prices always come from the dhaba's live menu, so they cannot be tampered with.

5

Pay

The selected seats are held for the passenger for 10 minutes while they pay, so nobody else can take them. If they do not finish paying in time, the seats are released for others automatically.

6

Get the ticket

The moment payment succeeds, the booking is confirmed and gets a PNR (a short ticket number like ZBK2G874MQ) and a QR code. The ticket also lists the seats, boarding and dropping points, bus number, driver contact and any food ordered.

7

Track the bus

One hour before their boarding time, the passenger can see the bus moving on a map. The position updates the moment the driver's phone sends it.

8

Board

The conductor scans the QR code (or types the PNR) at the door. The ticket is marked "on board". A ticket for another bus or another day is refused.

9

Eat on the way

At the dhaba stop, the food is already cooked and waiting.

IF PLANS CHANGE

The passenger can cancel all or some seats from "My bookings". The refund follows the operator's cancellation rules (for example 90% back if cancelled 48 hours before). Food that the dhaba has not started cooking is refunded in full. The freed seats go back on sale everywhere straight away.

SECTION 3

Seats: why one seat is never sold twice

A bus is often sold in several places at once: the ZI Bus app, the operator's own counter, agents, and other booking platforms. The biggest risk in this business is selling the same seat to two people. ZI Bus is built so that this cannot happen.

The "one seat, one lock" rule

Think of every seat on every trip as having a single lock. Before anyone can sell a seat, from anywhere, they must take its lock. The database lets exactly one seller take it; everyone else is told "already taken" and is shown the seats that are still free. In testing, 20 sellers tried to grab the same seat at the same instant and exactly one succeeded.

- **Holding:** while a passenger is paying, the lock is held for 10 minutes. Unpaid holds are released automatically, even if nobody is looking at that bus.
- **Selling:** once paid, the lock becomes permanent for that trip and the ticket gets its PNR.
- **Cancelling:** the lock is removed and the seat is back on sale everywhere at once.

Accurate counts at any moment

For every trip the admin panel shows: total seats, booked, being paid for right now, blocked by the operator, and available. It also shows who sold each booked seat (ZI Bus app, operator counter, or a named platform) and the full passenger list, which can be printed as the trip manifest.

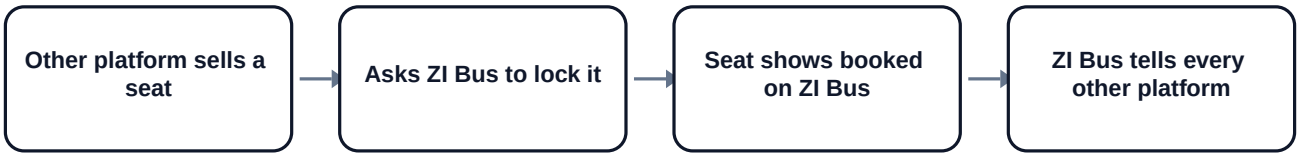
SECTION 3 CONTINUED

Working with other booking platforms

Example from the brief: a bus has 40 seats and 15 were sold on RedBus. Those 15 must show as booked on ZI Bus, and anything sold on ZI Bus must show as booked on RedBus. ZI Bus supports this in two ways, depending on what the other side can do.

Way 1: they connect to ZI Bus

A platform or agent system that can talk to us gets its own secure key. Before selling a seat, it asks ZI Bus to lock it; if the seat is free, the lock is theirs. If not, they are told which seats are taken. Every change that happens on ZI Bus (our own sales, cancellations, sales by other platforms) is sent back to them instantly, with a digital signature so they know it really came from us. If their system is down, ZI Bus keeps retrying for about half an hour.



Way 2: ZI Bus reads the operator's own system

Many bus companies already use reservation software that also feeds RedBus and others. If that software cannot call us, ZI Bus calls it instead: every couple of minutes it reads the bookings for each linked trip over the next week and makes ZI Bus match. Seats sold there are locked here; seats cancelled there are freed here. The ZI Bus team sets this up in the admin panel by filling in where their system keeps the booking number, the seats and the status, and which of their trips is which ZI Bus bus. Seat names that differ (for example their "U-7" is our "7A") can be matched too.

> **IF THE SAME SEAT WAS SOLD IN BOTH PLACES**
>
> ZI Bus never silently overwrites a sale. If the operator's system sold a seat that ZI Bus had already sold, it is flagged in the admin panel as a double-sale conflict so the operator can contact one of the passengers. With Way 1 this cannot happen at all; with Way 2 it can only happen in the minutes between two checks.

Before connecting a specific system (for example Bitla or Vaagai), we need that company's API details. Most will fit the ready-made connector by filling in a form; any that do not can get a small dedicated connector.

SECTION 4

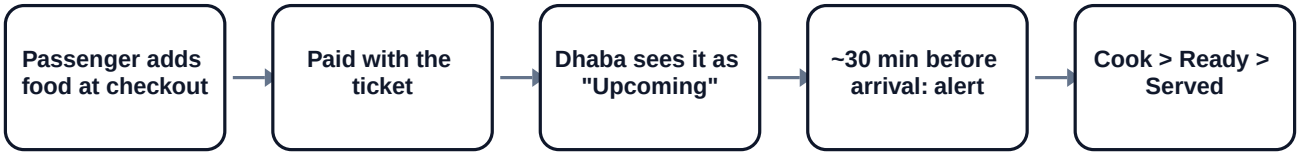
Dhaba food: from menu to plate

Partner dhabas are highway restaurants where ZI Bus buses stop. Each dhaba has its own login to a simple panel that works on any phone or computer.

Setting a dhaba up (done once by the ZI Bus team)

- Create the dhaba with its name, location on the map, login mobile number and password, and the ZI Bus commission on food.
- Link it to the buses that stop there, with how many minutes after departure the bus reaches it (for example "Pune to Nagpur night bus reaches here 5 hours after leaving") and how long it halts.
- The dhaba adds its own menu: dish name, price, veg or non-veg, and whether it is available today. Marking a dish "sold out" hides it from passengers immediately.

How an order flows



1. **Order placed**
   When the passenger's payment succeeds, the order appears in the dhaba's "Upcoming" list with the bus, passenger count, seats and dishes. It is already paid; the dhaba collects nothing at the counter.

2. **The 30-minute alert**
   About 30 minutes before the bus is expected, the dhaba's panel plays a chime, shows a pop-up and moves the order to "Prepare now". A phone notification is also sent, and an SMS when that is switched on. Expected arrival starts from the timetable; once the bus is moving, its live GPS position is used, so if the bus is running early the alert comes early too.

3. **Cooking**
   The dhaba taps Accept, Start cooking, Mark ready and finally Served to bus. Every device logged in for that dhaba stays in step.

4. **If the dhaba cannot make it**
   The dhaba taps "Can't make" and gives a reason. The passenger is notified and refunded for that food automatically.

WHAT THE ALERT TELLS THE DHABA

Bus name and number, expected arrival time, number of passengers, seat numbers, passenger name, the full list of dishes with quantities, and the bus crew's phone number.

Dhaba earnings

The dhaba panel shows orders and sales for today and the dhaba's payout for the month (food sales minus the ZI Bus commission).

SECTION 5

Bus operators and bus crew

Bus companies

Each bus company is set up once in ZI Bus with its brand name (what passengers see), legal name, GST number, contact details and default commission. One or more login accounts are attached to it, and its buses are assigned to it.

- **Buses, routes and timings:** operators add buses with photos, amenities, seat layout (seater, sleeper, lower and upper deck), route and stops, schedules for each day of the week, fares by seat type, and cancellation rules.
- **Bookings and passengers:** operators see every booking and the passenger list for each trip, and can block seats or make counter bookings.
- **Settlement report:** for any date range, ZI Bus shows each bus's seats sold (split by where they were sold), ticket sales made on ZI Bus, ZI Bus commission and the amount payable to the operator. A commission set on an individual bus overrides the company default.

Bus crew

- **Start the trip:** the driver starts the journey in the driver app. From then on the bus position is shared with passengers who have a ticket and are within their tracking window.
- **Check tickets at the door:** the "Check tickets" screen shows the trip's passenger list with seats, boarding points and who has boarded. The conductor scans the passenger's QR code with the phone camera or types the PNR. Counts of seats sold, on board and still to board update as they go.
- **Wrong tickets are refused:** a ticket for a different bus, a different trip or a different date is rejected with a clear message. Scanning the same ticket twice does no harm.

SECTION 6

The ZI Bus team and the money

What the admin panel controls

Page	Used for
Fleet Manager / Bus Bookings	Create and approve buses, see and manage all bookings, make manual bookings, cancel seats.
Bus Operators / Bus Commission	Bus companies, their logins and buses, their settlement reports, and commission per company or per bus.
Seat Inventory	Live seat counts for any trip, who sold each seat, printable passenger manifest.
Booking Channels	Connect other platforms and operator software, issue their keys, see every sync and any double-sale conflicts.
Dhaba Partners	Add dhabas, link them to buses and stops, set their commission, turn them on or off.
Dhaba Food Orders	Every food order with its status, plus food sales, ZI Bus commission and any refunds that need attention.

Who pays what

Event	What happens to the money
Passenger books	One online payment covers the seats (fare plus tax) and any food.
Seat sold on ZI Bus	ZI Bus keeps its commission; the rest is owed to the bus operator (see the settlement report).
Seat sold on another platform	That platform handles the money with the operator. ZI Bus only keeps the seat count correct.
Food order	ZI Bus keeps the dhaba commission; the rest is owed to the dhaba.
Passenger cancels	Seats are refunded by the operator's cancellation rules. Food not yet being cooked is refunded in full.
Dhaba cannot make the food	That food is refunded in full automatically.
Paid, but seats lost while paying	The whole payment is refunded automatically.

SECTION 7

Where things stand

Built and tested

- Search, live seat maps, seat holding and the "one seat, one lock" rule (tested with 20 people buying the same seat at once).
- Connections to other platforms, both ways described in Section 3, including automatic retries and conflict flagging.
- PNR and QR tickets, boarding and dropping points, conductor check-in.
- Dhaba panel, menus, food at checkout, the 30-minute alert adjusted by live GPS, order steps and refunds.
- Live bus tracking from one hour before boarding.
- Bus companies, settlement reports and all admin pages listed in Section 6.

Needed before launch

- **Live payment keys:** payments and refunds are built but must be tested with real Razorpay keys on a test server.
- **Notifications:** phone notifications need Firebase keys; dhaba SMS needs a message template approved under India's DLT rules.
- **Operator systems:** API details from each operator's reservation software before connecting it.
- **Test run:** a full trial on a staging server with a real bus, a real dhaba and a few test passengers.

Possible next improvements

- Dhaba arrival times based on road routes instead of straight-line distance.
- Separate staff roles inside a bus company (manager, booking clerk, conductor).
- A downloadable PDF ticket in addition to the in-app ticket.

WORDS USED IN THIS GUIDE

Glossary

Word	Meaning
PNR	The passenger's ticket number, for example ZBK2G874MQ. Given the moment payment succeeds. Letters that look alike (O and 0, I and 1) are never used.
QR code	The square barcode on the ticket. The conductor scans it to check the passenger in.
Trip / run	One bus on one schedule on one date, for example the Pune to Nagpur 21:00 bus on 21 October.
Seat hold	A seat kept aside for 10 minutes while a passenger pays.
Channel	Any other place that sells ZI Bus seats: another booking platform, an agent, or the operator's own counter or software.
Connector	The part of ZI Bus that reads bookings from an operator's software on a timer.
Webhook	An instant message ZI Bus sends to another platform whenever seats change.
Double-sale conflict	A seat that two places sold. ZI Bus flags it instead of overwriting either sale.
Manifest	The passenger list for one trip.
Settlement	What ZI Bus owes a bus company or dhaba after commission.
DLT	India's registration system for business SMS. Every SMS template must be approved before it can be sent.